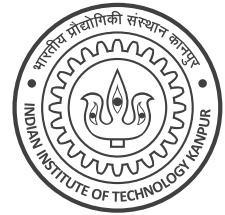


CS 771A: Intro to Machine Learning, IIT Kanpur			Quiz II (02 Apr 2026)	
Name	MELBO			20 marks Page 1 of 2
Roll No	260007	Dept.	AWSM	

Instructions:

1. This question paper contains 1 page (2 sides of paper). Please verify.
2. Write your name, roll number, department above in **block letters neatly with ink**.
3. Write your final answers neatly **with a blue/black pen**. Pencil marks may get smudged.
4. Don't overwrite/scratch answers especially in MCQ – such cases may get straight 0 marks.
5. Do not rush to fill in answers. You have enough time to solve this quiz.



Q1 (Machine Unlearning) Melbo learnt a model $\mathbf{w} \in \mathbb{R}^d$ (hidden bias) via least squares regression on N points $X \in \mathbb{R}^{N \times d}, \mathbf{y} \in \mathbb{R}^N$ with $\mathbf{x}^i \in \mathbb{R}^d, y^i \in \mathbb{R}, i \in [N]$. Derive an expression for $\mathbf{w}$ assuming $X^T X$ is invertible. Melba filed a DPDP complaint asking the j^{th} data point (containing Melba's personal data) to be removed from model training. Melbo, being lazy, did not remove $\mathbf{x}^j$ from the dataset and instead, set $y^j = 0$ and learnt a new model $\tilde{\mathbf{w}}$ using least squares on those same N points. Derive an expression for $\mathbf{w} - \tilde{\mathbf{w}}$. Simplify it as much as possible. **(2 + 4 = 6 marks)**

Least squares regression solves $\min_{\mathbf{w}} \frac{1}{2} \|X\mathbf{w} - \mathbf{y}\|_2^2$. Since this is an unconstrained problem with a differentiable objective, first-order optimality tells us $X^T(X\mathbf{w} - \mathbf{y}) = \mathbf{0}$ i.e. $\mathbf{w} = (X^T X)^{-1} X^T \mathbf{y}$. Melbo's lazy *unlearning* step essentially uses a different label vector, namely $\mathbf{y} - y^j \cdot \mathbf{e}_j$ where $\mathbf{e}_j \in \mathbb{R}^N$ is the j^{th} canonical one-hot basis vector with 1 at the j^{th} location and 0 everywhere else. Thus, a similar argument as above tells us that $\tilde{\mathbf{w}} = (X^T X)^{-1} X^T (\mathbf{y} - y^j \cdot \mathbf{e}_j)$. This tells us that

$$\mathbf{w} - \tilde{\mathbf{w}} = (X^T X)^{-1} X^T (y^j \cdot \mathbf{e}_j) = (X^T X)^{-1} (y^j \cdot \mathbf{x}^j)$$

Note: this lazy form of unlearning adopted by Melbo is easy to perform. When a DPDP removal request comes, say for the k^{th} point, find the diff vector $\Delta = (X^T X)^{-1} (y^k \cdot \mathbf{x}^k)$ and the *unlearned* model will simply be $\mathbf{w} - \Delta$. No need to solve least squares all over again.

Q2 (IPL Intrigue). Melbo is a big IPL fan and is trying to analyse the performance of KKR vs SRH on various kinds of pitches. Let K be the event that KKR won a KKR-vs-SRH match and S be the event that SRH won a KKR-vs-SRH match. There are 3 kinds of pitches $F = \text{flat}, G = \text{green}, D = \text{dusty}$. A total of 32 KKR-vs-SRH matches were played, 37.5% of which were on dusty pitches. Equal number of matches were played on flat and green pitches. The two teams are equally matched i.e. $\mathbb{P}[K] = \mathbb{P}[S]$. On flat pitches, KKR has a 4 $\times$ chance of winning i.e. $\mathbb{P}[K | F] = 4 \cdot \mathbb{P}[S | F]$. This advantage flips exactly on green pitches i.e. $\mathbb{P}[K | G] = \mathbb{P}[S | G]$. Find out the following quantities as fractions or decimals (no derivations needed). *Hint: either use Bayes rule or fill-up a 2 x 3 matrix showing which team won how many matches on what kind of pitch.* **(8 x 1 = 8 marks)**

$$\mathbb{P}[K | D] = 1/2$$

$$\mathbb{P}[S | D] = 1/2$$

$$\mathbb{P}[F | K] = 1/2$$

$$\mathbb{P}[F | S] = 1/8$$

$$\mathbb{P}[G | K] = 1/8$$

$$\mathbb{P}[G | S] = 1/2$$

$$\mathbb{P}[D | K] = 3/8$$

$$\mathbb{P}[D | S] = 3/8$$

Q3 (Gaussian Regularization?) Consider a data point $\mathbf{x} \in \mathbb{R}^d$, $y \in \mathbb{R}$, linear model $\mathbf{w} \in \mathbb{R}^d$ (hidden bias), a noise vector $\mathbf{z} \in \mathbb{R}^d \sim \mathcal{N}(\mathbf{0}, I_d)$ sampled from a standard normal distribution, and a noisy version of the least squares loss $\ell(\mathbf{w}, \mathbf{z}) \stackrel{\text{def}}{=} \frac{1}{2} (y - \mathbf{w}^\top (\mathbf{x} + \mathbf{z}))^2$. Write down an expression for the gradient of the noisy least squares loss i.e., $\nabla_{\mathbf{w}} \ell(\mathbf{w}, \mathbf{z})$ (no derivation needed). Then, derive the expected gradient $\mathbb{E}_{\mathbf{z}}[\nabla_{\mathbf{w}} \ell(\mathbf{w}, \mathbf{z})]$. *Note: the gradient is w.r.t. the model $\mathbf{w}$ while the expectation is w.r.t. the noise term $\mathbf{z}$.* Simplify your expressions as much as possible. Note that $\mathbb{E}_{\mathbf{z}}[\nabla_{\mathbf{w}} \ell(\mathbf{w}, \mathbf{z})]$ shouldn't contain any $\mathbf{z}$ terms although $\nabla_{\mathbf{w}} \ell(\mathbf{w}, \mathbf{z})$ will contain $\mathbf{z}$ terms. (2 + 4 = 6 marks)

$$\nabla_{\mathbf{w}} \ell(\mathbf{w}, \mathbf{z}) = (\mathbf{x} + \mathbf{z})((\mathbf{x} + \mathbf{z})^\top \mathbf{w} - y)$$

Write final expression in the box

$$\mathbb{E}_{\mathbf{z}}[\nabla_{\mathbf{w}} \ell(\mathbf{w}, \mathbf{z})] = \mathbf{w} + \mathbf{x} \cdot (\mathbf{x}^\top \mathbf{w} - y)$$

Give brief derivation with simplification below

$$\begin{aligned} \mathbb{E}_{\mathbf{z}}[\nabla_{\mathbf{w}} \ell(\mathbf{w}, \mathbf{z})] &= \mathbb{E}_{\mathbf{z}}[(\mathbf{x} + \mathbf{z})((\mathbf{x} + \mathbf{z})^\top \mathbf{w} - y)] \\ &= \mathbb{E}_{\mathbf{z}}[(\mathbf{xx}^\top + \mathbf{xz}^\top + \mathbf{zx}^\top + \mathbf{zz}^\top)\mathbf{w} - y \cdot (\mathbf{x} + \mathbf{z})] \\ &= (\mathbb{E}_{\mathbf{z}}[\mathbf{xx}^\top] + \mathbf{x}\mathbb{E}_{\mathbf{z}}[\mathbf{z}]^\top + \mathbb{E}_{\mathbf{z}}[\mathbf{z}]\mathbf{x}^\top + \mathbb{E}_{\mathbf{z}}[\mathbf{zz}^\top])\mathbf{w} - y \cdot (\mathbf{x} + \mathbb{E}_{\mathbf{z}}[\mathbf{z}]) \\ &= (\mathbf{xx}^\top + I_d)\mathbf{w} - y \cdot \mathbf{x} = \mathbf{w} + \mathbf{x} \cdot (\mathbf{x}^\top \mathbf{w} - y) \end{aligned}$$

where in the third step we used linearity of expectation, in the fourth step, we used the fact that $\mathbb{E}_{\mathbf{z}}[\mathbf{z}] = \mathbf{0}$, $\mathbb{E}_{\mathbf{z}}[\mathbf{xx}^\top] = \mathbf{xx}^\top$ since $\mathbf{z}$ is sampled independently, and $\mathbb{E}_{\mathbf{z}}[\mathbf{zz}^\top] = I_d$ since that is the covariance of the standard normal distribution.

Note that the expected gradient is identical to that of ridge regression even though we used no explicit regularization in the loss function. This is regularization by adding noise.